

application development, and a new chain-based Byzantine fault-tolerant consensus algorithm called Sumeragi. The most defining characteristic of Hyperledger Iroha is its ability to be freely interoperable with other Hyperledger projects. The open source libraries for iOS, Android and JavaScript allow developers to conveniently create functions for performing common operations.

Figure 1 shows a layered architectural view of the different components that make up Hyperledger Iroha. The four layers are: API, Peer Interaction, Chain Business Logic and Storage.

Here is a brief description of the components of Hyperledger Iroha.

Model classes are system entities.

Torii (gate) provides the input and output interfaces for clients. It is a single gRPC server that is used by clients to interact with peers through the network. The client's RPC call is non-blocking, making *Torii* an asynchronous server.

Both commands (transactions) and queries (read access) are performed through this interface.

Network encompasses interaction with the network of peers.

Consensus is in charge of peers agreeing on chain content in the network. The consensus mechanism used by Iroha is YAC (Yet Another Consensus), which is a practical Byzantine fault-tolerant algorithm based on voting for block hash.

Simulator generates a temporary snapshot of storage to validate transactions by executing them against this snapshot and forming a verified proposal, which consists only of valid transactions.

Validator classes check the business rules and validity (correct format) of transactions or queries.

There are two distinct types of validation that occur in Hyperledger Iroha:

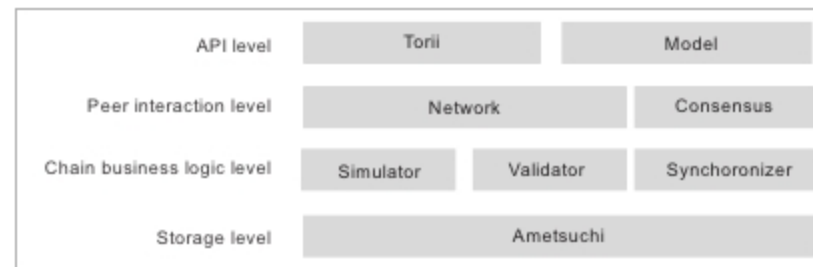


Figure 1: Hyperledger Iroha architecture



Figure 2: Transaction flow: Step 1

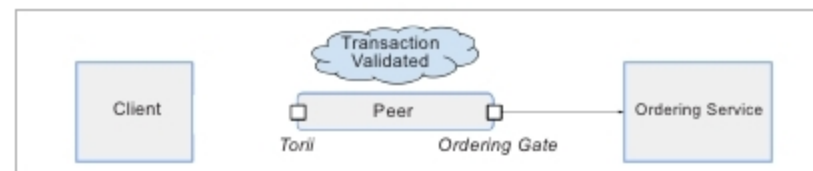


Figure 3: Transaction flow: Step 2

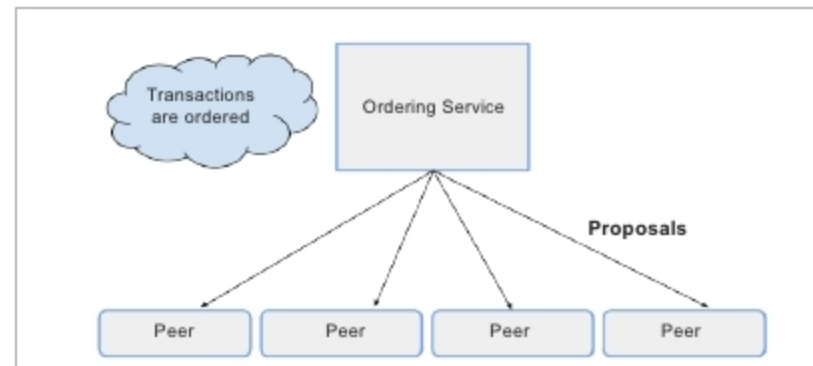


Figure 4: Transaction flow: Step 3

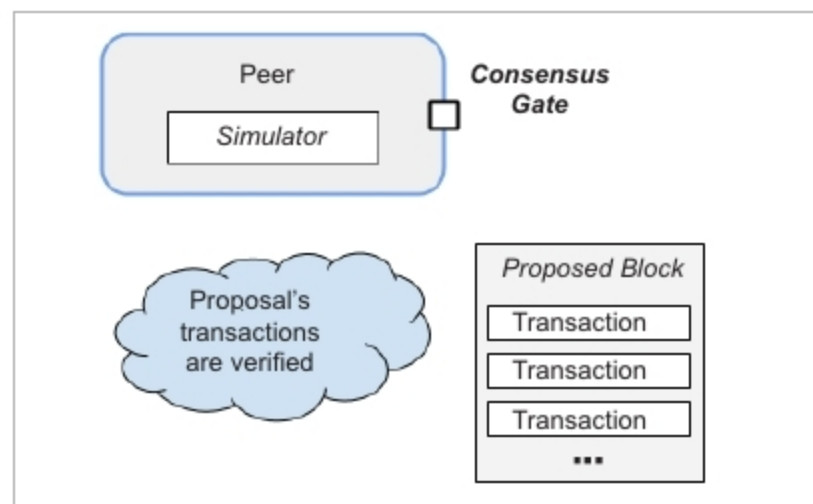


Figure 5: Transaction flow: Step 4

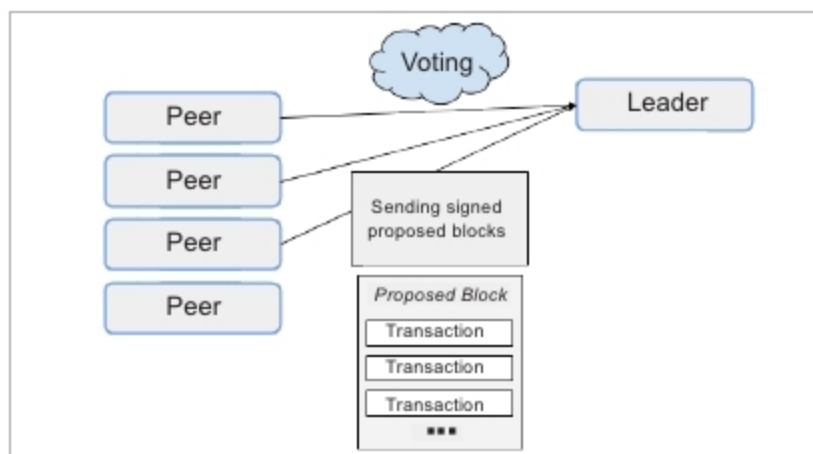


Figure 6: Transaction flow: Step 5

- Stateless validation is a quicker form of validation, and it performs schema and signature checks of the transaction.

- Stateful validation is slower and checks the permissions and the current world state view (which is the latest and most actual state of the chain) to see if desired business rules and policies are possible — for example, to check if an account has enough funds to transfer.

Synchroniser helps to synchronise new peers in the system or temporarily disconnect peers.

Ametsuchi is the ledger block storage which consists of a block index, block store, and a world state view component.

Key participants in the network

There are three main participants in a Hyperledger Iroha network.

Clients are able to:

- Query data that they have access/permission to;
- Perform a state-changing action or 'transaction', which consists of atomic operations, called 'commands'.

Peers maintain the current state and their own copy of the shared ledger. A peer is a single entity in the network, and has an address, identity and trust.

The *ordering service* orders transactions into a known order. There are a few options for the algorithm used by the ordering service.

Transaction flow

Step 1: A client creates and sends a transaction to the *Torii* gate, which routes the transaction to a peer that is responsible for performing stateless validation.